

Atomic Structure

MCQ

1. All matter is made up of tiny particles called
 - a) cells
 - b) molecules
 - c) atoms
 - d) tissues

Answer: c) atoms

2. Atoms of the same element have the same
 - a) colour
 - b) size and mass
 - c) shape and charge
 - d) position

Answer: b) size and mass

3. The basic unit of an element is
 - a) molecule
 - b) compound
 - c) atom
 - d) mixture

Answer: c) atom

4. An atom is electrically neutral because
 - a) it has no particles
 - b) it contains neutrons only
 - c) positive and negative charges are equal
 - d) electrons are absent

Answer: c) positive and negative charges are equal

5. Molecules are formed by the combination of
 - a) elements only
 - b) atoms
 - c) compounds
 - d) mixtures

Answer: b) atoms

6. The centre of an atom is called

- a) shell
- b) orbit
- c) nucleus
- d) molecule

Answer: c) nucleus

7. Which particle has a positive charge?

- a) Electron
- b) Neutron
- c) Proton
- d) Atom

Answer: c) Proton

8. Which particle revolves around the nucleus?

- a) Proton
- b) Neutron
- c) Electron
- d) Molecule

Answer: c) Electron

9. Neutrons are

- a) positively charged
- b) negatively charged
- c) neutral
- d) unstable

Answer: c) neutral

10. Atoms cannot be seen with the naked eye because they are

- a) colourless
- b) very small
- c) transparent
- d) inactive

Answer: b) very small

11. The approximate diameter of an atom is

- a) 1 m

- b) 1 mm
- c) 1×10^{-9} m
- d) 1 km

Answer: c) 1×10^{-9} m

12. Who proposed the first atomic theory?

- a) J.J. Thomson
- b) Rutherford
- c) Dalton
- d) Chadwick

Answer: c) Dalton

13. Dalton proposed his atomic theory in the year

- a) 1708
- b) 1808
- c) 1897
- d) 1908

Answer: b) 1808

14. According to Dalton, atoms are

- a) destructible
- b) divisible
- c) indivisible
- d) radioactive

Answer: c) indivisible

15. Dalton's theory could not explain

- a) size of atoms
- b) shape of atoms
- c) electrical charges
- d) mass of atoms

Answer: c) electrical charges

16. J.J. Thomson discovered

- a) proton
- b) neutron
- c) electron

d) nucleus

Answer: c) electron

17. Thomson's atomic model is known as

a) nuclear model

b) planetary model

c) plum pudding model

d) solid sphere model

Answer: c) plum pudding model

18. In Thomson's model, electrons are

a) outside the atom

b) embedded in positive charge

c) inside the nucleus

d) absent

Answer: b) embedded in positive charge

19. Rutherford conducted his experiment using

a) silver foil

b) copper wire

c) gold foil

d) iron sheet

Answer: c) gold foil

20. The rays used in Rutherford's experiment were

a) beta rays

b) gamma rays

c) alpha rays

d) X-rays

Answer: c) alpha rays

21. Most alpha particles passed through the gold foil showing that

a) atom is solid

b) atom has no charge

c) atom has empty space

d) atom is heavy

Answer: c) atom has empty space

22. The nucleus is
- a) negatively charged
 - b) neutral
 - c) positively charged
 - d) empty

Answer: c) positively charged

23. Most of the mass of an atom is concentrated in the
- a) electrons
 - b) orbits
 - c) nucleus
 - d) shells

Answer: c) nucleus

24. The size of the nucleus is
- a) equal to atom
 - b) larger than atom
 - c) very small compared to atom
 - d) variable

Answer: c) very small compared to atom

25. Subatomic particles are
- a) larger than atoms
 - b) smaller than atoms
 - c) equal to atoms
 - d) molecules

Answer: b) smaller than atoms

26. Which particle was discovered by James Chadwick?
- a) Electron
 - b) Proton
 - c) Neutron
 - d) Nucleus

Answer: c) Neutron

27. Protons and neutrons together are called
- a) electrons

- b) isotopes
- c) nucleons
- d) orbits

Answer: c) nucleons

28. Atomic number is equal to the number of

- a) neutrons
- b) protons
- c) nucleons
- d) shells

Answer: b) protons

29. Atomic number is represented by the letter

- a) A
- b) N
- c) Z
- d) X

Answer: c) Z

30. Mass number is the total number of

- a) electrons and protons
- b) neutrons only
- c) protons and neutrons
- d) electrons only

Answer: c) protons and neutrons

31. Mass number is represented by the letter

- a) Z
- b) A
- c) X
- d) N

Answer: b) A

32. Isotopes have the same

- a) mass number
- b) atomic number
- c) number of neutrons

d) valency

Answer: b) atomic number

33. Isobars have the same

a) atomic number

b) electrons

c) mass number

d) valency

Answer: c) mass number

34. Valency refers to

a) mass of atom

b) size of atom

c) combining capacity

d) charge of atom

Answer: c) combining capacity

35. Valency depends on the number of electrons in the

a) nucleus

b) second shell

c) outermost shell

d) innermost shell

Answer: c) outermost shell

36. An atom becomes positively charged when it

a) gains electrons

b) loses electrons

c) gains neutrons

d) loses neutrons

Answer: b) loses electrons

37. An atom becomes negatively charged when it

a) loses electrons

b) loses neutrons

c) gains electrons

d) gains protons

Answer: c) gains electrons

38. An element with valency one is called

- a) divalent
- b) trivalent
- c) monovalent
- d) multivalent

Answer: c) monovalent

39. Which of the following is a monovalent element?

- a) Oxygen
- b) Nitrogen
- c) Sodium
- d) Aluminium

Answer: c) Sodium

40. Elements with valency two are called

- a) monovalent
- b) divalent
- c) trivalent
- d) tetravalent

Answer: b) divalent

41. Oxygen is an example of a

- a) monovalent element
- b) divalent element
- c) trivalent element
- d) noble gas

Answer: b) divalent element

42. Trivalent elements have valency

- a) one
- b) two
- c) three
- d) four

Answer: c) three

43. Aluminium is a

- a) monovalent element

- b) divalent element
- c) trivalent element
- d) inert element

Answer: c) trivalent element

44. Variable valency means

- a) fixed charge
- b) no charge
- c) more than one valency
- d) no electrons

Answer: c) more than one valency

45. Iron shows variable valency of

- a) 1 and 2
- b) 2 and 3
- c) 3 and 4
- d) 1 and 3

Answer: b) 2 and 3

46. Ferrous oxide contains iron with valency

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b) 2

47. Ferric oxide contains iron with valency

- a) 1
- b) 2
- c) 3
- d) 4

Answer: c) 3

48. The symbol of an electron is

- a) p
- b) n
- c) e

d) a

Answer: c) e

49. The symbol used for neutron is

a) e

b) p

c) n

d) z

Answer: c) n

50. Nuclear notation gives information about

a) colour and size

b) atomic and mass number

c) valency only

d) charge only

Answer: b) atomic and mass number